# OEE Tracker

Step-by-step guide: track Overall Equipment Effectiveness by machine and shift

## What it does

You log one row per machine per shift. The workbook calculates Availability, Performance, Quality and OEE for every row, then summarizes each machine on the **Dashboard** for any date range you choose, with a chart and a comparison to your target.

| | |
|---|---|
| Yellow cells: type here. | Grey cells: calculated. Don't type over them. |

## The sheets

| Sheet | What it's for |
|---|---|
| How to use | A one-page summary. |
| Setup | Your machines and each one's ideal cycle time. Up to 20 machines. |
| Production Log | One row per machine per shift. Up to 500 rows. |
| Dashboard | Totals and OEE by machine for a date range, plus a chart. |

## Before you start

1. Open the file in your spreadsheet app (see **Works with** in the listing). If Excel shows a yellow *Protected View* bar because the file was downloaded, click **Enable Editing**.
2. Save a copy under your own name (for example, add your plant or year) and keep the original download as a clean spare.
3. Read the **How to use** sheet. It's a one-page summary of this guide.

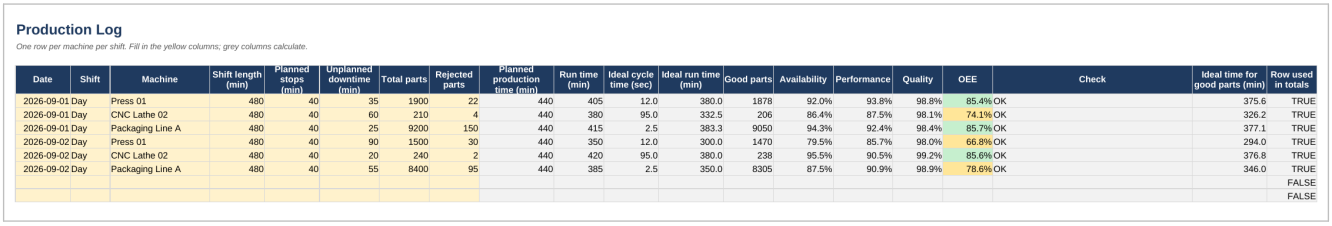
## Step 1: Set up your machines

1. Go to the **Setup** sheet.
2. In **Machine**, type each machine's name once, one per row (up to 20). Use the names your team already uses.
3. In **Ideal cycle time (seconds per part)**, enter the fastest time the machine can sustainably make one good part. For a line, use the time per finished unit at rated speed. Example: a machine rated at 300 parts per hour has an ideal cycle time of 3600 / 300 = **12 seconds**.
4. Type each machine only once. If a name is repeated, **Check** on the repeat says **Duplicate: only the first row is used**; delete the repeat.
5. Clear the three EXAMPLE machines when you no longer need them (yellow cells only).

Getting the ideal cycle time right matters most: if it's too slow, Performance can show over 100% and the Check column will flag it.

## Step 2: Log each shift

1. Go to the **Production Log** sheet.
2. Fill in one row per machine per shift. The yellow columns are: **Date**, **Shift** (any label, e.g. Day or 1st), **Machine** (pick from the drop-down), **Shift length (min)**, **Planned stops (min)**, **Unplanned downtime (min)**, **Total parts** and **Rejected parts**. Shift length and total parts are required; blank planned stops, downtime or rejects count as 0.
3. **Planned stops** are times the machine isn't expected to run, like breaks and scheduled meetings. **Unplanned downtime** is every other minute it was stopped: breakdowns, changeovers, waiting for material.
4. The grey columns fill in by themselves, including **OEE** (coloured green, amber or red) and **Check**.
5. Look at **Check** on each new row. It should say **OK**. If not, see the table below. Rows with an error are left out of the Dashboard until you fix them. The "performance above 100%" warning is not an error: those rows still count.
6. The rows marked **EXAMPLE** show the expected format. When you're ready, select just their **yellow** cells and press **Delete** to clear them. Don't delete whole rows: that removes the formulas in the grey columns too.



**Production Log**

*One row per machine per shift. Fill in the yellow columns; grey columns calculate.*

| Date | Shift | Machine | Shift length (min) | Planned stops (min) | Unplanned downtime (min) | Total parts | Rejected parts | Planned production time (min) | Run time (min) | Ideal cycle time (sec) | Ideal run time (min) | Good parts | Availability | Performance | Quality | OEE | Check | Ideal time for good parts (min) | Row used in totals |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2026-09-01 | Day | Press 01 | 480 | 40 | 35 | 1900 | 22 | 440 | 405 | 12.0 | 380.0 | 1878 | 92.0% | 93.8% | 98.8% | 85.4% | OK | 375.6 | TRUE |
| 2026-09-01 | Day | CNC Lathe 02 | 480 | 40 | 60 | 210 | 4 | 440 | 380 | 95.0 | 332.5 | 206 | 86.4% | 87.5% | 98.1% | 74.1% | OK | 326.2 | TRUE |
| 2026-09-01 | Day | Packaging Line A | 480 | 40 | 25 | 9200 | 150 | 440 | 415 | 2.5 | 383.3 | 9050 | 94.3% | 92.4% | 98.4% | 85.7% | OK | 377.1 | TRUE |
| 2026-09-02 | Day | Press 01 | 480 | 40 | 90 | 1500 | 30 | 440 | 350 | 12.0 | 300.0 | 1470 | 79.5% | 85.7% | 98.0% | 66.8% | OK | 294.0 | TRUE |
| 2026-09-02 | Day | CNC Lathe 02 | 480 | 40 | 20 | 240 | 2 | 440 | 420 | 95.0 | 380.0 | 238 | 95.5% | 90.5% | 99.2% | 85.6% | OK | 376.8 | TRUE |
| 2026-09-02 | Day | Packaging Line A | 480 | 40 | 55 | 8400 | 95 | 440 | 385 | 2.5 | 350.0 | 8305 | 87.5% | 90.9% | 98.9% | 78.6% | OK | 346.0 | TRUE |
| | | | | | | | | | | | | | | | | | | | FALSE |
| | | | | | | | | | | | | | | | | | | | FALSE |

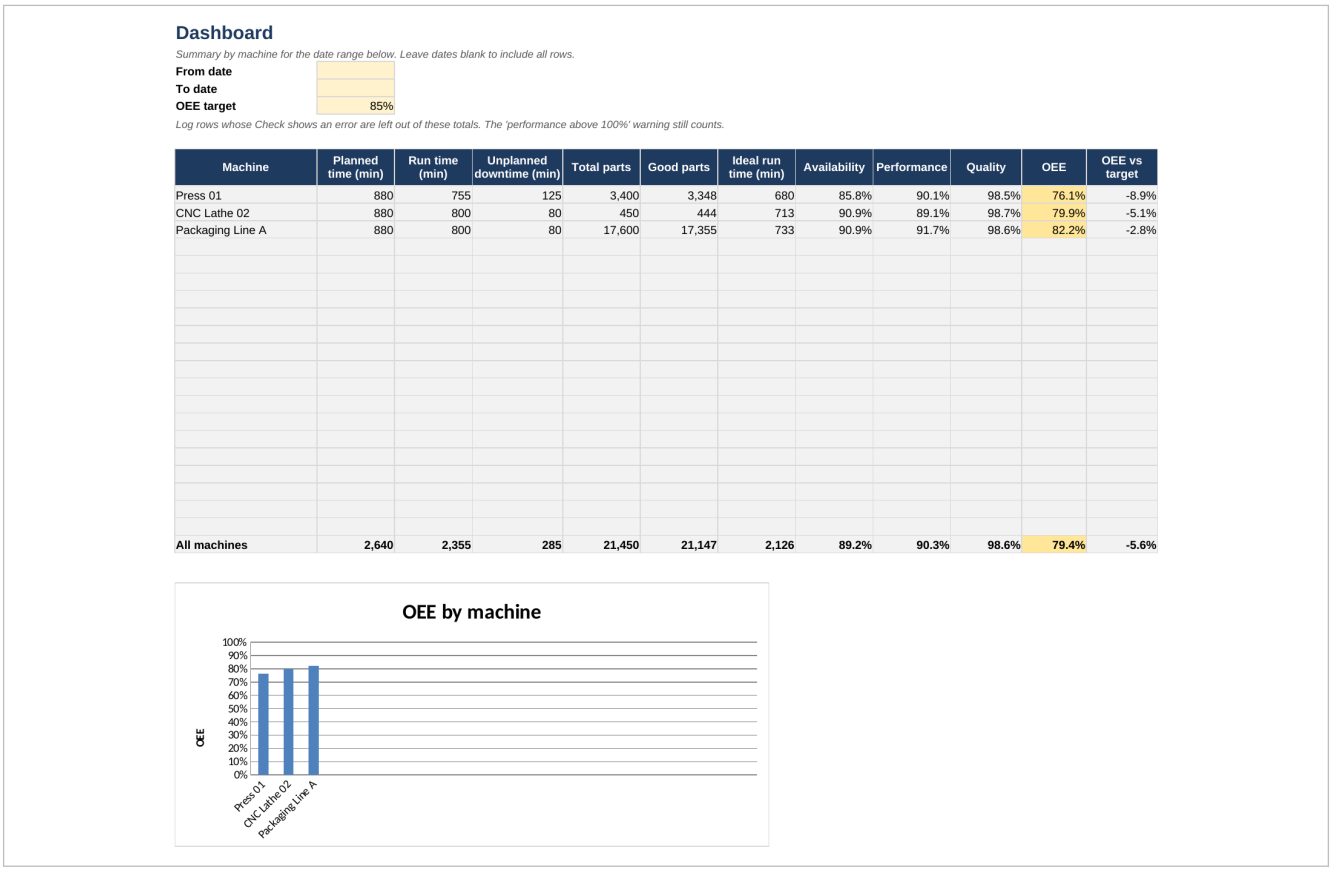
Production Log with the example rows. Yellow columns are typed in; grey columns calculate.

### What the Check column means

| Message | What to do |
|---|---|
| Machine not in Setup | The machine name isn't on the Setup sheet, or is spelled differently. Pick it from the drop-down, or add it to Setup. |
| Set an ideal cycle time in Setup | The machine has no ideal cycle time (or 0) on the Setup sheet. |
| Enter a date | The Date cell is empty or isn't a real date. Undated rows can't be placed in a date range, so they are left out. |
| Enter numbers: shift length and total parts are required | A required cell is empty, or a cell contains text instead of a number. |
| Values must be 0 or more (shift length above 0) | A negative number, or a shift length of 0. |
| Rejects > total parts | Rejected parts must be part of the total. Fix one of the two numbers. |
| Planned stops >= shift length | Planned stops leave no time to run. Check both numbers. |
| Downtime > planned time | Unplanned downtime is longer than shift length minus planned stops. Check the three time columns. |
| Check: performance above 100% | A warning only; the row still counts. The machine made parts faster than its ideal cycle time allows. Usually the cycle time in Setup is too slow, or the count is wrong. |
| OK | Nothing obviously wrong. |

## Step 3: Read the Dashboard

1. Go to the **Dashboard** sheet.
2. Optional: enter a **From date** and/or **To date**. Leave them blank to include every row. Both dates are included in the range.
3. Set your **OEE target** (85% is filled in). OEE cells turn **green** at or above target, **amber** from 60% up to the target, and **red** below 60%. Red always wins: below 60% is red even if your target is lower than 60%. **OEE vs target** shows the gap.
4. Each row adds up times and part counts first, then calculates the percentages, so a long shift counts more than a short one. **All machines** OEE is total ideal time for good parts divided by total planned production time, which stays correct when machines have different cycle times. Its **Quality** is weighted by ideal time (ideal time for good parts ÷ ideal time for all parts), so Availability × Performance × Quality still equals OEE.



**Dashboard**

*Summary by machine for the date range below. Leave dates blank to include all rows.*

**From date**
**To date**
**OEE target** 85%

*Log rows whose Check shows an error are left out of these totals. The 'performance above 100%' warning still counts.*

| Machine | Planned time (min) | Run time (min) | Unplanned downtime (min) | Total parts | Good parts | Ideal run time (min) | Availability | Performance | Quality | OEE | OEE vs target |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Press 01 | 880 | 755 | 125 | 3,400 | 3,348 | 680 | 85.8% | 90.1% | 98.5% | 76.1% | -8.9% |
| CNC Lathe 02 | 880 | 800 | 80 | 450 | 444 | 713 | 90.9% | 89.1% | 98.7% | 79.9% | -5.1% |
| Packaging Line A | 880 | 800 | 80 | 17,600 | 17,355 | 733 | 90.9% | 91.7% | 98.6% | 82.2% | -2.8% |
| **All machines** | **2,640** | **2,355** | **285** | **21,450** | **21,147** | **2,126** | **89.2%** | **90.3%** | **98.6%** | **79.4%** | **-5.6%** |



Dashboard: totals and OEE by machine, the all-machines total, and the OEE chart.

## How the numbers are calculated

| Measure | Formula |
|---|---|
| Planned production time | Shift length - planned stops |
| Run time | Planned production time - unplanned downtime |
| Availability | Run time ÷ planned production time |
| Performance | (Ideal cycle time × total parts) ÷ run time |
| Quality | Good parts ÷ total parts (good parts = total - rejected) |
| OEE | Availability × Performance × Quality = (ideal cycle time × good parts) ÷ planned production time |

85% OEE is a commonly cited "world-class" benchmark. Many plants start well below it; improving against your own baseline matters more.

## Questions

**The log is full (500 rows). What now?** Save the workbook as a new file for the next period (for example, one per quarter), clear the yellow cells in the log, and keep going.

**Can I add more than 20 machines?** Not in this version. Message us if you need more.

**Can I rename the sheets or move columns?** Please don't. The formulas refer to them by position.

## Need help?

Message us through Etsy (the **Message seller** button on our shop or your order). Tell us which file, which sheet, and what you typed. We answer questions about using the templates as described in this guide; we can't do custom changes or set-up work.

These templates were designed with help from AI tools and checked by the seller.

# Downtime Log & Pareto

Step-by-step guide: log every stop and see which causes cost the most time

## What it does

You record each stop with a reason. The **Pareto** sheet ranks the reasons by minutes lost, with each reason's share and the running total, so you can see the few causes behind most of your downtime. Filter by date range and machine.

| | |
|---|---|
| Yellow cells: type here. | Grey cells: calculated. Don't type over them. |

## The sheets

| Sheet | What it's for |
|---|---|
| How to use | A one-page summary. |
| Setup | Your machines (up to 30) and your downtime reasons with categories (up to 30). |
| Downtime Log | One row per stop. Up to 1,000 rows. |
| Pareto | Reasons ranked by minutes lost, with a Pareto chart of the top 12. |
| By Machine | Downtime minutes and stop counts per machine. |

## Before you start

1. Open the file in your spreadsheet app (see **Works with** in the listing). If Excel shows a yellow *Protected View* bar because the file was downloaded, click **Enable Editing**.
2. Save a copy under your own name (for example, add your plant or year) and keep the original download as a clean spare.
3. Read the **How to use** sheet. It's a one-page summary of this guide.

## Step 1: Set up machines and reasons

1. Go to the **Setup** sheet.
2. Under **Machine**, list your machines, one per row.
3. Under **Downtime reason** and **Category**, edit the 12 starter reasons to match how your plant talks about stops. Keep the list short, about 10 to 15 reasons. Categories (Breakdown, Material, and so on) can be renamed too.
4. List each machine and each reason only once. A repeated name shows **Duplicate: only the first row is used** in the **Check** column next to it; delete the repeat.

**Setup**

*Machines and downtime reasons used by the drop-downs.*

| Machine | Check |
|---|---|
| Press 01 | |
| CNC Lathe 02 | |
| Packaging Line A | |
| | |
| | |
| | |
| | |
| | |
| | |
| | |
| | |
| | |
| | |
| | |

| Downtime reason | Category | Check |
|---|---|---|
| Mechanical breakdown | Breakdown | |
| Electrical / controls fault | Breakdown | |
| Tooling failure | Breakdown | |
| Changeover / setup | Setup & adjustment | |
| Adjustment / tuning | Setup & adjustment | |
| Waiting for material | Material | |
| Waiting for upstream process | Material | |
| Quality hold / inspection | Quality | |
| No operator available | Labor | |
| Planned maintenance | Planned | |
| Meeting / training | Planned | |
| Other (describe in notes) | Other | |
| | | |
| | | |

Setup: machines on the left, reasons and categories on the right.

## Step 2: Log every stop

1. Go to the **Downtime Log** sheet and use one row per stop.
2. Enter **Date**, **Shift** and **Machine** (from the drop-down).
3. Enter **Start** and **End** as times, like **7:15** and **7:50**. A stop that crosses midnight, like 23:40 to 0:25, is handled automatically (45 minutes).
4. If you only know how long it lasted, leave Start and End blank and type the minutes in **Minutes (if no start/end)**. Typed minutes always win over start and end times.
5. Pick a **Reason** from the drop-down. **Duration**, **Category** and **Check** fill in by themselves. Check should say **OK**; otherwise it tells you what's missing, such as a date, a reason, or minutes that aren't a number of 0 or more.
6. Add **Notes** if useful, especially for "Other".
7. The rows marked **EXAMPLE** show the expected format. When you're ready, select just their **yellow** cells and press **Delete** to clear them. Don't delete whole rows: that removes the formulas in the grey columns too.

**Downtime Log**

*One row per stop. Use start/end times, or type minutes if you only know the duration.*

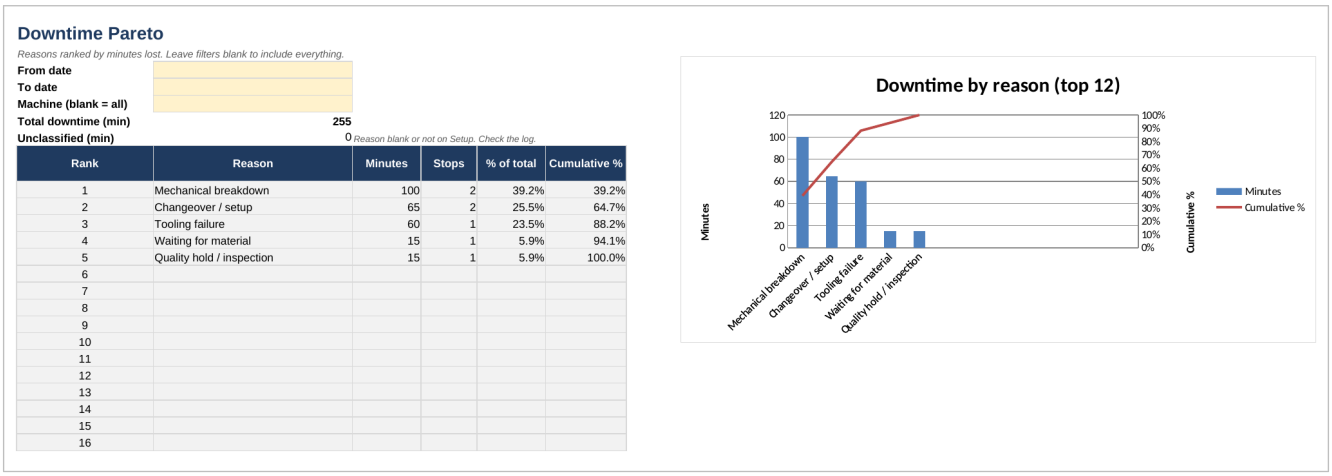
| Date | Shift | Machine | Start (hh:mm) | End (hh:mm) | Minutes (if no start/end) | Duration (min) | Reason | Category | Notes | Check |
|---|---|---|---|---|---|---|---|---|---|---|
| 2026-09-01 | Day | Press 01 | 07:15 | 07:50 | | 35 | Changeover / setup | Setup & adjustment | EXAMPLE - Die change | OK |
| 2026-09-01 | Day | Press 01 | 10:05 | 10:20 | | 15 | Waiting for material | Material | EXAMPLE | OK |
| 2026-09-01 | Day | CNC Lathe 02 | 09:00 | 10:00 | | 60 | Tooling failure | Breakdown | EXAMPLE - Insert broke | OK |
| 2026-09-01 | Night | Packaging Line A | 23:40 | 00:25 | | 45 | Mechanical breakdown | Breakdown | EXAMPLE - Conveyor jam, crossed midnight | OK |
| 2026-09-02 | Day | Press 01 | | | 55 | 55 | Mechanical breakdown | Breakdown | EXAMPLE - Minutes typed directly | OK |
| 2026-09-02 | Day | Packaging Line A | 13:00 | 13:30 | | 30 | Changeover / setup | Setup & adjustment | EXAMPLE | OK |
| 2026-09-02 | Day | CNC Lathe 02 | 14:10 | 14:25 | | 15 | Quality hold / inspection | Quality | EXAMPLE | OK |
| | | | | | | | | | | |
| | | | | | | | | | | |

Downtime Log with the example rows, including a stop that crosses midnight.

If **Category** shows **Not in Setup**, the reason was typed instead of picked, or is spelled differently from the Setup list. Its minutes still count in the total, under **Unclassified** on the Pareto sheet. Pick the reason from the drop-down so it's counted under the right reason. A row counts as a **stop** when its duration is more than 0 minutes; rows without a date, or with invalid minutes, are left out of the totals.

## Step 3: Read the Pareto

1. Go to the **Pareto** sheet.
2. Optional: set **From date**, **To date** and **Machine**. Leave any of them blank to include everything. Both dates are included in the range.
3. **Total downtime** is every valid logged minute in the range. **Unclassified** shows minutes whose reason is blank or not on the Setup list. Fix those in the log so they're ranked.
4. Read the table from the top: **Rank**, **Reason**, **Minutes**, **Stops**, **% of total** and **Cumulative %**. Reasons with no minutes in the range are left out.
5. The chart shows the top 12 reasons as bars, with the cumulative % as a line on the right-hand axis.
6. Start improvement work with the reasons at the top. Often the first two or three account for most of the lost time.



**Downtime Pareto**

*Reasons ranked by minutes lost. Leave filters blank to include everything.*

| | |
|---|---|
| From date | |
| To date | |
| Machine (blank = all) | |
| Total downtime (min) | 255 |
| Unclassified (min) | 0 *Reason blank or not on Setup. Check the log.* |

| Rank | Reason | Minutes | Stops | % of total | Cumulative % |
|---|---|---|---|---|---|
| 1 | Mechanical breakdown | 100 | 2 | 39.2% | 39.2% |
| 2 | Changeover / setup | 65 | 2 | 25.5% | 64.7% |
| 3 | Tooling failure | 60 | 1 | 23.5% | 88.2% |
| 4 | Waiting for material | 15 | 1 | 5.9% | 94.1% |
| 5 | Quality hold / inspection | 15 | 1 | 5.9% | 100.0% |
| 6 | | | | | |
| 7 | | | | | |
| 8 | | | | | |
| 9 | | | | | |
| 10 | | | | | |
| 11 | | | | | |
| 12 | | | | | |
| 13 | | | | | |
| 14 | | | | | |
| 15 | | | | | |
| 16 | | | | | |



Pareto: reasons ranked by minutes, with cumulative % and the Pareto chart.

## Step 4: Compare machines

The **By Machine** sheet shows total downtime, number of stops, average minutes per stop and each machine's share of all downtime, plus **No machine / not in Setup** and **All logged downtime** rows so the totals always add up. It uses the date range set on the Pareto sheet; the machine filter doesn't apply here.

## Downtime by Machine

*Uses the date range set on the Pareto sheet (machine filter is ignored here). A stop is any row with more than 0 min*

| Machine | Downtime (min) | Stops | Avg min per stop | % of all downtime |
|---|---|---|---|---|
| Press 01 | 105 | 3 | 35.0 | 41.2% |
| CNC Lathe 02 | 75 | 2 | 37.5 | 29.4% |
| Packaging Line A | 75 | 2 | 37.5 | 29.4% |
| | | | | |
| | | | | |

By Machine: downtime minutes, stops and share per machine.

## Tips

- If **Other** climbs into your top reasons, read its notes and add the missing reason to Setup.
- Durations are rounded to whole minutes.
- Log stops the same day, while people remember the cause.

## Questions

**The log is full (1,000 rows). What now?** Save the workbook as a new file for the next period, clear the yellow cells in the log, and keep going.

**Can I rename the sheets or move columns?** Please don't. The formulas refer to them by position.

## Need help?

Message us through Etsy (the **Message seller** button on our shop or your order). Tell us which file, which sheet, and what you typed. We answer questions about using the templates as described in this guide; we can't do custom changes or set-up work.

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# Preventive Maintenance Schedule

Step-by-step guide: plan recurring maintenance and see what's due

## What it does

You list each recurring maintenance task and how often it's due. Each time a task is done, you add a line to the **Completion Log**. The plan then shows each task's last completion, next due date, days left and a colour-coded status.

| | |
|---|---|
| Yellow cells: type here. | Grey cells: calculated. Don't type over them. |

## The sheets

| Sheet | What it's for |
|---|---|
| How to use | A one-page summary. |
| PM Plan | One row per asset and task, with status. Up to 300 tasks. |
| Completion Log | One row each time a task is done. Up to 2,000 rows. |

## Before you start

1. Open the file in your spreadsheet app (see **Works with** in the listing). If Excel shows a yellow *Protected View* bar because the file was downloaded, click **Enable Editing**.
2. Save a copy under your own name (for example, add your plant or year) and keep the original download as a clean spare.
3. Read the **How to use** sheet. It's a one-page summary of this guide.

## Step 1: Build your plan

1. Go to the **PM Plan** sheet.
2. Add one row per asset and task. Fill in **Asset**, **Task**, **Frequency (days)** and **Assigned to**. The frequency is a fixed number of days: 1 = daily, 7 = weekly, 14 = every two weeks, 30 = about monthly, 90 = about quarterly, 365 = about yearly.
3. If you know when a task was last done before you started using this workbook, enter that date in **Last done before log**. Otherwise leave it blank: the task shows as due today, every day, until you log it or enter a date.
4. Keep each asset + task combination on one row only. A repeated combination shows **Duplicate row**; a blank, zero or negative frequency shows **Check frequency**. Both are counted under **Needs fixing** at the top right.

5 The rows marked **EXAMPLE** show the expected format. When you're ready, select just their **yellow** cells and press **Delete** to clear them. Don't delete whole rows: that removes the formulas in the grey columns too.

**PM Plan**

*One row per asset and task.*

**Due soon window (days)** 7

| Overdue | 1 |
|---|---|
| Due soon | 2 |
| OK | 2 |
| Needs fixi | 0 |

| Asset | Task | Frequency (days) | Assigned to | Last done before log | Last completed | Next due | Days until due | Status | Notes |
|---|---|---|---|---|---|---|---|---|---|
| Air compressor 1 | Change oil filter | 90 | Maintenance | 2026-07-02 | 2026-07-02 | 2026-09-30 | -10 | OVERDUE | EXAMPLE |
| Press 01 | Lubricate slides and gibs | 7 | Operator | 2026-10-07 | 2026-10-07 | 2026-10-14 | 4 | Due soon | EXAMPLE |
| Forklift 2 | Inspect forks, chains and hydraulics | 30 | Maintenance | 2026-09-13 | 2026-09-13 | 2026-10-13 | 3 | Due soon | EXAMPLE |
| CNC Lathe 02 | Clean chip conveyor and coolant filter | 14 | Operator | | 2026-10-08 | 2026-10-22 | 12 | OK | EXAMPLE |
| Dust collector | Replace filter cartridges | 180 | Maintenance | 2026-09-20 | 2026-09-20 | 2027-03-19 | 160 | OK | EXAMPLE |

PM Plan with example tasks, statuses and the counts box (top right).

## Step 2: Log completed work

1 Go to the **Completion Log** sheet.

2 Each time a task is done, add a row: **Date done**, **Asset** and **Task** (pick both from the drop-downs so they match the plan exactly), **Done by** and any **Notes**.

3 Check **Matches plan?**. It should say **OK**. **Enter the date done** means the date is missing. **Fill in asset and task** means one of them is blank. **No matching asset + task in plan** means the asset or task is spelled differently from the PM Plan, so the plan won't update. Fix the spelling.

**Completion Log**

*Add a row each time a task is completed. Pick asset and task from the drop-downs.*

| Date done | Asset | Task | Done by | Notes | Matches plan? |
|---|---|---|---|---|---|
| 2026-10-08 | CNC Lathe 02 | Clean chip conveyor and coolant filter | J. Smith | EXAMPLE | OK |

Completion Log: each completion updates the plan.

## Step 3: Read the plan

1 Back on **PM Plan**, the grey columns update automatically: **Last completed** (the most recent of the logged dates and the before-log date), **Next due**, **Days until due** and **Status**.

2 **OVERDUE** (red) means the due date has passed. **Due soon** (amber) means it's due within the window. **OK** (green) means it's not due yet.

3 Change the **Due soon window (days)** at the top of the sheet. It starts at 7.

4 The counts box at the top right shows how many tasks are Overdue, Due soon, OK, and Needs fixing.

5 To get this week's work list, sort or filter by Status: **Data > Filter** in Excel.

Dates and statuses use today's date from your computer, so they change from day to day. The example rows are set relative to today for the same reason.

## Questions

**What if a task is done early?** Log the actual date. The next due date counts from the last completion.

**The Completion Log is full.** Save the workbook as a new file, enter each task's most recent date in **Last done before log**, clear the yellow cells in the log, and keep going.

**Can I rename the sheets or move columns?** Please don't. The formulas refer to them by position.

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